

9.2 GPIO registers

Refer to [Section 2.2 on page 45](#) for a list of abbreviations used in register descriptions.

The peripheral registers have to be accessed by words (32-bit).

9.2.1 Port configuration register low (GPIOx_CRL) (x=A..G)

Address offset: 0x00

Reset value: 0x4444 4444

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CNF7[1:0]		MODE7[1:0]		CNF6[1:0]		MODE6[1:0]		CNF5[1:0]		MODE5[1:0]		CNF4[1:0]		MODE4[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNF3[1:0]		MODE3[1:0]		CNF2[1:0]		MODE2[1:0]		CNF1[1:0]		MODE1[1:0]		CNF0[1:0]		MODE0[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:30, 27:26, **CNFy[1:0]**: Port x configuration bits (y= 0 .. 7)
 23:22, 19:18, 15:14, 11:10, 7:6, 3:2 These bits are written by software to configure the corresponding I/O port.
 Refer to [Table 20: Port bit configuration table](#).

In input mode (MODE[1:0]=00):

00: Analog mode
 01: Floating input (reset state)
 10: Input with pull-up / pull-down
 11: Reserved

In output mode (MODE[1:0] > 00):

00: General purpose output push-pull
 01: General purpose output Open-drain
 10: Alternate function output Push-pull
 11: Alternate function output Open-drain

Bits 29:28, 25:24, **MODEy[1:0]**: Port x mode bits (y= 0 .. 7)
 21:20, 17:16, 13:12, 9:8, 5:4, 1:0 These bits are written by software to configure the corresponding I/O port.
 Refer to [Table 20: Port bit configuration table](#).

00: Input mode (reset state)
 01: Output mode, max speed 10 MHz.
 10: Output mode, max speed 2 MHz.
 11: Output mode, max speed 50 MHz.

9.2.2 Port configuration register high (GPIOx_CRH) (x=A..G)

Address offset: 0x04

Reset value: 0x4444 4444

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CNF15[1:0]		MODE15[1:0]		CNF14[1:0]		MODE14[1:0]		CNF13[1:0]		MODE13[1:0]		CNF12[1:0]		MODE12[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNF11[1:0]		MODE11[1:0]		CNF10[1:0]		MODE10[1:0]		CNF9[1:0]		MODE9[1:0]		CNF8[1:0]		MODE8[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:30, 27:26, **CNFy[1:0]**: Port x configuration bits (y= 8 .. 15)
 23:22, 19:18, 15:14, 11:10, 7:6, 3:2 These bits are written by software to configure the corresponding I/O port.
 Refer to [Table 20: Port bit configuration table](#).

In input mode (MODE[1:0]=00):

00: Analog mode
 01: Floating input (reset state)
 10: Input with pull-up / pull-down
 11: Reserved

In output mode (MODE[1:0] > 00):

00: General purpose output push-pull
 01: General purpose output Open-drain
 10: Alternate function output Push-pull
 11: Alternate function output Open-drain

Bits 29:28, 25:24, **MODEy[1:0]**: Port x mode bits (y= 8 .. 15)
 21:20, 17:16, 13:12, 9:8, 5:4, 1:0 These bits are written by software to configure the corresponding I/O port.
 Refer to [Table 20: Port bit configuration table](#).

00: Input mode (reset state)
 01: Output mode, max speed 10 MHz.
 10: Output mode, max speed 2 MHz.
 11: Output mode, max speed 50 MHz.

9.2.3 Port input data register (GPIOx_IDR) (x=A..G)

Address offset: 0x08h

Reset value: 0x0000 XXXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IDR15	IDR14	IDR13	IDR12	IDR11	IDR10	IDR9	IDR8	IDR7	IDR6	IDR5	IDR4	IDR3	IDR2	IDR1	IDR0
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **IDRy**: Port input data (y= 0 .. 15)

These bits are read only and can be accessed in Word mode only. They contain the input value of the corresponding I/O port.

9.2.4 Port output data register (GPIOx_ODR) (x=A..G)

Address offset: 0x0C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ODR15	ODR14	ODR13	ODR12	ODR11	ODR10	ODR9	ODR8	ODR7	ODR6	ODR5	ODR4	ODR3	ODR2	ODR1	ODR0
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **ODRy**: Port output data (y= 0 .. 15)

These bits can be read and written by software and can be accessed in Word mode only.

Note: For atomic bit set/reset, the ODR bits can be individually set and cleared by writing to the GPIOx_BSRR register (x = A .. G).

9.2.5 Port bit set/reset register (GPIOx_BSRR) (x=A..G)

Address offset: 0x10

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BR15	BR14	BR13	BR12	BR11	BR10	BR9	BR8	BR7	BR6	BR5	BR4	BR3	BR2	BR1	BR0
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BS15	BS14	BS13	BS12	BS11	BS10	BS9	BS8	BS7	BS6	BS5	BS4	BS3	BS2	BS1	BS0
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w

Bits 31:16 **BRy**: Port x Reset bit y (y= 0 .. 15)

These bits are write-only and can be accessed in Word mode only.

0: No action on the corresponding ODRx bit

1: Reset the corresponding ODRx bit

Note: If both BSx and BRx are set, BSx has priority.

Bits 15:0 **BSy**: Port x Set bit y (y= 0 .. 15)

These bits are write-only and can be accessed in Word mode only.

0: No action on the corresponding ODRx bit

1: Set the corresponding ODRx bit

9.2.6 Port bit reset register (GPIOx_BRR) (x=A..G)

Address offset: 0x14

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BR15	BR14	BR13	BR12	BR11	BR10	BR9	BR8	BR7	BR6	BR5	BR4	BR3	BR2	BR1	BR0
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w

Bits 31:16 Reserved

Bits 15:0 **BRy**: Port x Reset bit y (y= 0 .. 15)

These bits are write-only and can be accessed in Word mode only.

0: No action on the corresponding ODRx bit

1: Reset the corresponding ODRx bit

9.2.7 Port configuration lock register (GPIOx_LCKR) (x=A..G)

This register is used to lock the configuration of the port bits when a correct write sequence is applied to bit 16 (LCKK). The value of bits [15:0] is used to lock the configuration of the GPIO. During the write sequence, the value of LCKR[15:0] must not change. When the LOCK sequence has been applied on a port bit it is no longer possible to modify the value of the port bit until the next reset.

Each lock bit freezes the corresponding 4 bits of the control register (CRL, CRH).

Address offset: 0x18

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															LCKK
															rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LCK15	LCK14	LCK13	LCK12	LCK11	LCK10	LCK9	LCK8	LCK7	LCK6	LCK5	LCK4	LCK3	LCK2	LCK1	LCK0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:17 Reserved

Bit 16 **LCKK[16]**: Lock key

This bit can be read anytime. It can only be modified using the Lock Key Writing Sequence.

0: Port configuration lock key not active

1: Port configuration lock key active. GPIOx_LCKR register is locked until the next reset.

LOCK key writing sequence:

Write 1

Write 0

Write 1

Read 0

Read 1 (this read is optional but confirms that the lock is active)

Note: During the LOCK Key Writing sequence, the value of LCK[15:0] must not change.

Any error in the lock sequence will abort the lock.

Bits 15:0 **LCKy**: Port x Lock bit y (y= 0 .. 15)

These bits are read write but can only be written when the LCKK bit is 0.

0: Port configuration not locked

1: Port configuration locked.

9.3 Alternate function I/O and debug configuration (AFIO)

To optimize the number of peripherals available for the 64-pin or the 100-pin or the 144-pin package, it is possible to remap some alternate functions to some other pins. This is achieved by software, by programming the [AF remap and debug I/O configuration register \(AFIO_MAPR\)](#). In this case, the alternate functions are no longer mapped to their original assignments.

9.3.1 Using OSC32_IN/OSC32_OUT pins as GPIO ports PC14/PC15

The LSE oscillator pins OSC32_IN and OSC32_OUT can be used as general-purpose I/O PC14 and PC15, respectively, when the LSE oscillator is off. The LSE has priority over the GP I/Os function.

Note: The PC14/PC15 GPIO functionality is lost when the 1.8 V domain is powered off (by entering standby mode) or when the backup domain is supplied by V_{BAT} (V_{DD} no more supplied). In this case the I/Os are set in analog mode.

Refer to the note on IO usage restrictions in [Section 5.1.2: Battery backup domain](#).

9.3.2 Using OSC_IN/OSC_OUT pins as GPIO ports PD0/PD1

The HSE oscillator pins OSC_IN/OSC_OUT can be used as general-purpose I/O PD0/PD1 by programming the PD01_REMAP bit in the [AF remap and debug I/O configuration register \(AFIO_MAPR\)](#).

This remap is available only on 36-, 48- and 64-pin packages (PD0 and PD1 are available on 100-pin and 144-pin packages, no need for remapping).

Note: The external interrupt/event function is not remapped. PD0 and PD1 cannot be used for external interrupt/event generation on 36-, 48- and 64-pin packages.